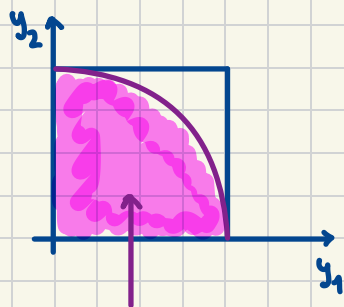
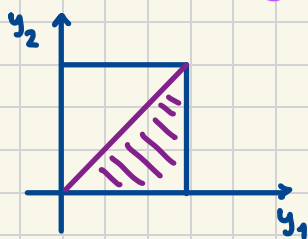


More on Random Vectors.

Example. $(Y_1, Y_2) \dots$ represents a point chosen
@ random in a unit square $[0,1]^2$

$$f_{Y_1, Y_2}(y_1, y_2) = \underset{\substack{= \\ \times \\ 1}}{c} \cdot \mathbb{1}_{[0,1] \times [0,1]}(y_1, y_2)$$

$$\mathbb{P}[Y_1 > Y_2] = \times \left(\frac{1}{2} \right) \times$$



$$A = \{(y_1, y_2) : y_1^2 + y_2^2 \leq 1\}$$

$$\mathbb{P}[Y_1^2 + Y_2^2 \leq 1] = \times \frac{\pi}{4}$$

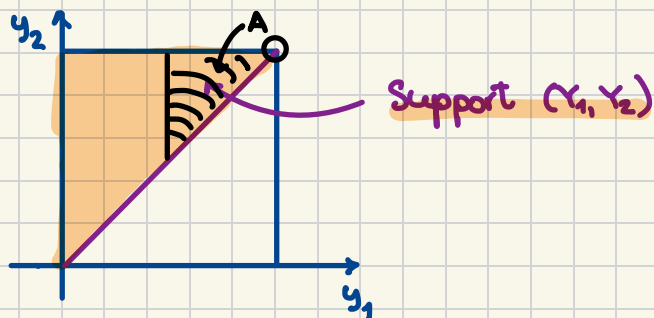
$$= \iint_A f_{Y_1, Y_2}(y_1, y_2) dy_2 dy_1 = \dots = \frac{\pi}{4}$$

$$\mathbb{P}[|Y_1 - Y_2| < \frac{1}{2}] = ?$$

Think on it @ home!

Example. Let (Y_1, Y_2) be jointly continuous w/ joint pdf

$$f_{Y_1, Y_2}(y_1, y_2) = 6y_1 \cdot \mathbb{1}_{[0 \leq y_1 \leq y_2 \leq 1]}$$



$$\mathbb{P}[Y_1 > \frac{1}{2}, Y_2 > \frac{1}{2}] = ?$$

$$= \iint_A f_{Y_1, Y_2}(y_1, y_2) dy_2 dy_1$$

$$= \int_{\frac{1}{2}}^1 \int_{\frac{1}{2}}^1 6y_1 \cdot \mathbb{1}_{[0 \leq y_1 \leq y_2 \leq 1]} dy_2 dy_1$$

$$= \int_{\frac{1}{2}}^1 6y_1 \int_{\frac{1}{2}}^1 \mathbb{1}_{[0 \leq y_1 \leq y_2 \leq 1]} dy_2 dy_1$$

$$= \int_{\frac{1}{2}}^1 6y_1 \int_{y_1}^1 dy_2 dy_1$$

$$= \int_{\frac{1}{2}}^1 6y_1(1-y_1) dy_1 = 6 \int_{\frac{1}{2}}^1 (y_1 - y_1^2) dy_1 = 6 \left(\frac{y_1^2}{2} - \frac{y_1^3}{3} \right) \Big|_{y_1=\frac{1}{2}}^1$$

$$= 6 \left(\frac{1}{2} - \frac{1}{3} - \frac{1}{8} + \frac{1}{24} \right) = 6 \cdot \frac{12-8-3+1}{24} = \frac{1}{2} \square$$

Functions of Random Vectors.

Theorem. Let $(Y_1, \dots, Y_n)$ be a jointly continuous r.v.
w/ the joint pdf

$$f_{Y_1, \dots, Y_n}(\dots\dots).$$

Let g be a real function of n variables
such that we can define

$$W = g(Y_1, \dots, Y_n)$$

Then,

$$\mathbb{E}[W] = \int \int \dots \int_{\mathbb{R}^n} g(y_1, \dots, y_n) f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) dy_n \dots dy_1$$

Example. [previous cont'd]

$$\mathbb{E}[Y_1^2 + Y_2^2] = X$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y_1^2 + y_2^2) 6y_1 \cdot \mathbb{1}_{[0 \leq y_1 \leq y_2 \leq 1]} dy_2 dy_1$$

$$= 6 \int_0^1 \int_{y_1}^1 (y_1^2 + y_2^2) y_1 dy_2 dy_1$$

$$= 6 \int_0^1 \int_{y_1}^1 (y_1^3 + y_1 y_2^2) dy_2 dy_1$$

$$= 6 \int_0^1 \left(y_1^3 y_2 + y_1 \cdot \frac{y_2^3}{3} \right) \Big|_{y_2=y_1}^1 dy_1$$

$$= 6 \int_0^1 \left(y_1^3 + \frac{y_1}{3} - y_1^4 - \frac{y_1^4}{3} \right) dy_1$$

$$= 6 \int_0^1 \left(y_1^3 + \frac{y_1}{3} - \frac{4}{3} y_1^4 \right) dy_1$$

$$= \int_0^1 (6y_1^3 + 2y_1 - 8y_1^4) dy_1$$

$$= \left. \frac{6y_1^4}{4} + \frac{2y_1^2}{2} - \frac{8y_1^5}{5} \right|_{y_1=0}^1 = \frac{3}{2} + 1 - \frac{8}{5} = \frac{9}{10} \quad \square$$